```
#Installing libraries
import pandas as pd
```

```
trxn= pd.read_csv('Retail_Data_Transactions.csv')
```

```
trxn
```

	customer_id	trans_date	tran_amount
0	CS5295	11-Feb-13	35
1	CS4768	15-Mar-15	39
2	CS2122	26-Feb-13	52
3	CS1217	16-Nov-11	99
4	CS1850	20-Nov-13	78
...	...	...	...
124995	CS8433	26-Jun-11	64
124996	CS7232	19-Aug-14	38
124997	CS8731	28-Nov-14	42
124998	CS8133	14-Dec-13	13
124999	CS7996	13-Dec-14	36

125000 rows × 3 columns

```
response= pd.read_csv('Retail_Data_Response.csv')
response
```

	customer_id	response
0	CS1112	0
1	CS1113	0
2	CS1114	1
3	CS1115	1
4	CS1116	1
...	...	...
6879	CS8996	0
6880	CS8997	0
6881	CS8998	0
6882	CS8999	0
6883	CS9000	0

6884 rows × 2 columns

```
df= trxn.merge(response, on='customer_id', how='left')
df
```

	customer_id	trans_date	tran_amount	response
0	CS5295	11-Feb-13	35	1.0
1	CS4768	15-Mar-15	39	1.0
2	CS2122	26-Feb-13	52	0.0
3	CS1217	16-Nov-11	99	0.0
4	CS1850	20-Nov-13	78	0.0
...	...	...	...	...
124995	CS8433	26-Jun-11	64	0.0
124996	CS7232	19-Aug-14	38	0.0
124997	CS8731	28-Nov-14	42	0.0
124998	CS8133	14-Dec-13	13	0.0
124999	CS7996	13-Dec-14	36	0.0

125000 rows × 4 columns

```
# features

df.dtypes
df.shape
df.tail()
```

	customer_id	trans_date	tran_amount	response
124995	CS8433	26-Jun-11	64	0.0
124996	CS7232	19-Aug-14	38	0.0
124997	CS8731	28-Nov-14	42	0.0
124998	CS8133	14-Dec-13	13	0.0
124999	CS7996	13-Dec-14	36	0.0

```
df.describe()
```

	tran_amount	response
count	125000.000000	124969.000000
mean	64.991912	0.110763
std	22.860006	0.313840
min	10.000000	0.000000
25%	47.000000	0.000000
50%	65.000000	0.000000
75%	83.000000	0.000000
max	105.000000	1.000000

```
# MISSING VALUES

df.isnull().sum()
```

	0
customer_id	0
trans_date	0
tran_amount	0
response	31

dtype: int64

```
df=df.dropna()
```

```
df
```

	customer_id	trans_date	tran_amount	response
0	CS5295	11-Feb-13	35	1.0
1	CS4768	15-Mar-15	39	1.0
2	CS2122	26-Feb-13	52	0.0
3	CS1217	16-Nov-11	99	0.0
4	CS1850	20-Nov-13	78	0.0
...	...	...	...	...
124995	CS8433	26-Jun-11	64	0.0
124996	CS7232	19-Aug-14	38	0.0
124997	CS8731	28-Nov-14	42	0.0
124998	CS8133	14-Dec-13	13	0.0
124999	CS7996	13-Dec-14	36	0.0

124969 rows × 4 columns

```
# change dtypes

df['trans_date']= pd.to_datetime(df['trans_date'])
df['response']= df['response'].astype('int64')
```

```
/tmp/ipykernel_15323/2474506332.py:3: UserWarning: Could not infer format, so each element will be parsed individually, falling
  df['trans_date']= pd.to_datetime(df['trans_date'])
/tmp/ipykernel_15323/2474506332.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-ver
  df['trans_date']= pd.to_datetime(df['trans_date'])
/tmp/ipykernel_15323/2474506332.py:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-ver
  df['response']= df['response'].astype('int64')
```

```
df
```

	customer_id	trans_date	tran_amount	response
0	CS5295	2013-02-11	35	1
1	CS4768	2015-03-15	39	1
2	CS2122	2013-02-26	52	0
3	CS1217	2011-11-16	99	0
4	CS1850	2013-11-20	78	0
...	...	...	...	...
124995	CS8433	2011-06-26	64	0
124996	CS7232	2014-08-19	38	0
124997	CS8731	2014-11-28	42	0
124998	CS8133	2013-12-14	13	0
124999	CS7996	2014-12-13	36	0

124969 rows × 4 columns

```
set(df['response'])
```

```
{0, 1}
```

```
df.dtypes
```

	0
customer_id	object
trans_date	datetime64[ns]
tran_amount	int64
response	int64

dtype: object

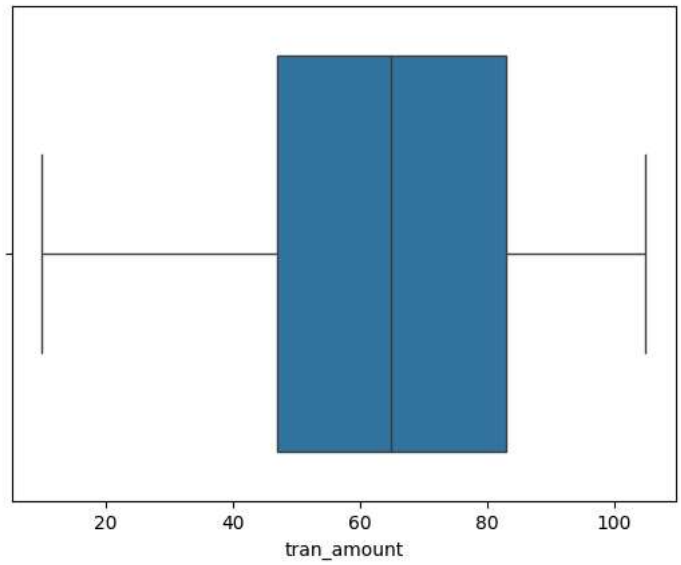
```
# check for outliers
#Z-SCORE

from scipy import stats
import numpy as np

#calc z score
z_scores= np.abs(stats.zscore(df['tran_amount']))

#set a threshold

threshold= 3

outliers= z_scores>threshold


print(df[outliers])
```

```
Empty DataFrame
Columns: [customer_id, trans_date, tran_amount, response]
Index: []
```

```
# check for outliers
#Z-SCORE

from scipy import stats
import numpy as np

#calc z score
z_scores= np.abs(stats.zscore(df['response']))

#set a threshold

threshold= 3

outliers= z_scores>threshold


print(df[outliers])
```

```
Empty DataFrame
Columns: [customer_id, trans_date, tran_amount, response]
Index: []
```

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(x=df['tran_amount'])
plt.show()
```




```
# creating new columns

df['month']= df['trans_date'].dt.month
```

```
/tmp/ipykernel_15323/197175934.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-ver
  df['month']= df['trans_date'].dt.month
```

```
df
```

	customer_id	trans_date	tran_amount	response	month
0	CS5295	2013-02-11	35	1	2
1	CS4768	2015-03-15	39	1	3
2	CS2122	2013-02-26	52	0	2
3	CS1217	2011-11-16	99	0	11
4	CS1850	2013-11-20	78	0	11
...	...	...	...	...	...
124995	CS8433	2011-06-26	64	0	6
124996	CS7232	2014-08-19	38	0	8
124997	CS8731	2014-11-28	42	0	11
124998	CS8133	2013-12-14	13	0	12
124999	CS7996	2014-12-13	36	0	12

124969 rows × 5 columns

```
# Which 3 months have had the highest transaction amounts?

monthly_Sales= df.groupby('month')['tran_amount'].sum()
monthly_Sales= monthly_Sales.sort_values(ascending=False).reset_index().head(3)
monthly_Sales
```

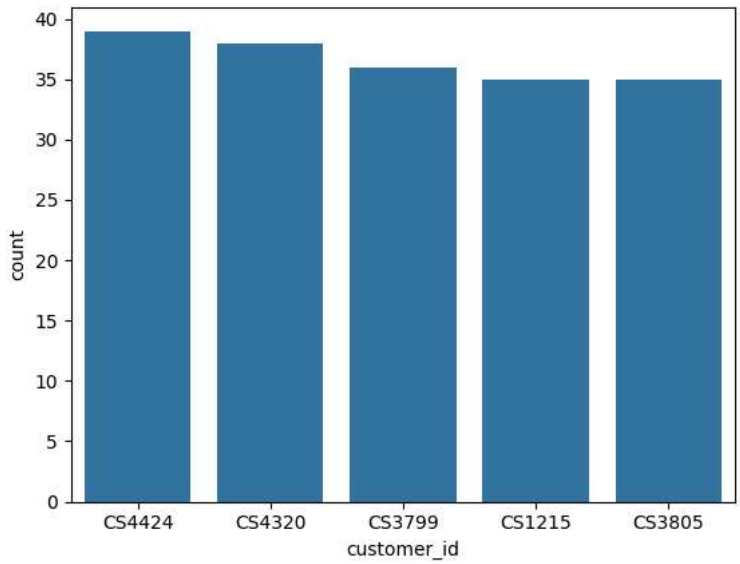
	month	tran_amount
0	8	726775
1	10	725058
2	1	724089

```
# Customers having highest num of orders

customer_counts= df['customer_id'].value_counts().reset_index()
customer_counts.columns=['customer_id','count']

# sort

top_5_cus= customer_counts.sort_values(by='count', ascending=False).head(5)
top_5_cus
```

	customer_id	count
0	CS4424	39
1	CS4320	38
2	CS3799	36
4	CS1215	35
3	CS3805	35

```
sns.barplot(x='customer_id',y='count',data=top_5_cus)
```

```
<Axes: xlabel='customer_id', ylabel='count'>
```

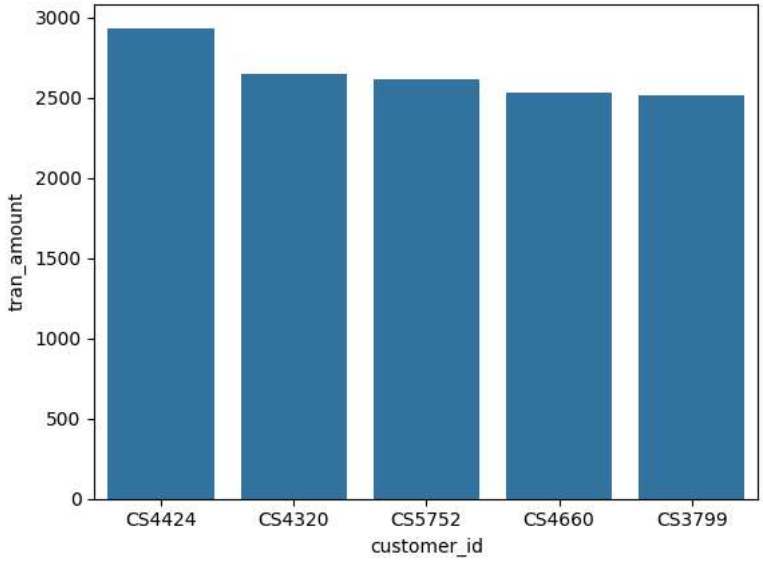



```
# Customers having highest value of orders

customer_sales= df.groupby('customer_id')['tran_amount'].sum().reset_index()
customer_sales

# sort

top_5_sal= customer_sales.sort_values(by='tran_amount', ascending=False).head(5)
top_5_sal
```

	customer_id	tran_amount
3312	CS4424	2933
3208	CS4320	2647
4640	CS5752	2612
3548	CS4660	2527
2687	CS3799	2513

```
sns.barplot(x='customer_id',y='tran_amount',data=top_5_sal)
```

```
<Axes: xlabel='customer_id', ylabel='tran_amount'>
```

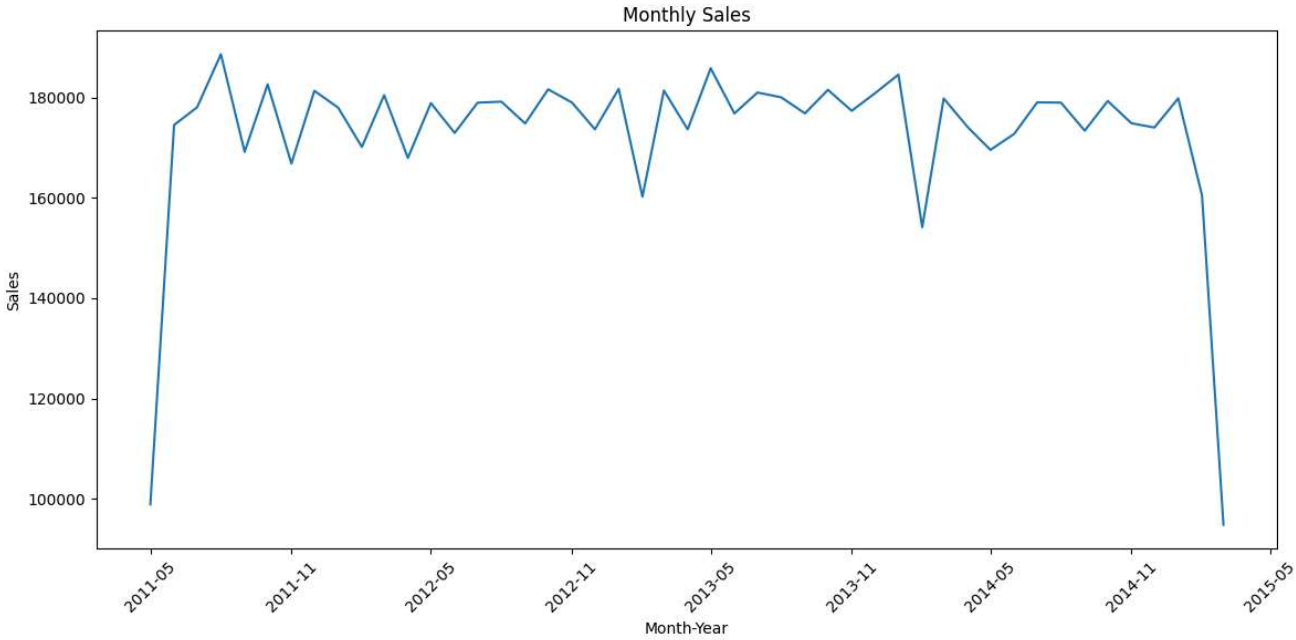


Advanced Analytics

Time Series Analysis

```
import matplotlib.pyplot as plt
import matplotlib.dates as mdates

df['month_year'] = df['trans_date'].dt.to_period('M')
monthly_sales = df.groupby('month_year')['tran_amount'].sum()

# Convert the PeriodIndex to DateTimeIndex
monthly_sales.index = monthly_sales.index.to_timestamp()

plt.figure(figsize=(12,6))  # Increase the size of the figure
plt.plot(monthly_sales.index, monthly_sales.values)  # Plot the data
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%Y-%m'))  # Format the x-axis labels
plt.gca().xaxis.set_major_locator(mdates.MonthLocator(interval=6))  # Set the x-axis interval
plt.xlabel('Month-Year')
plt.ylabel('Sales')
plt.title('Monthly Sales')
plt.xticks(rotation=45)  # Rotate the x-axis labels
plt.tight_layout()  # Adjust the layout for better visibility
plt.show()
```

```
/tmp/ipykernel_15323/3171706351.py:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-ver
  df['month_year'] = df['trans_date'].dt.to_period('M')
```



Cohort Segmentation

```
# Recency will be the maximum of trans_date
recency = df.groupby('customer_id')['trans_date'].max()

# Frequency will be the count of transactions
frequency = df.groupby('customer_id')['trans_date'].count()

# Monetary will be the sum of tran_amount
monetary = df.groupby('customer_id')['tran_amount'].sum()

# Combine all three into a DataFrame
rfm = pd.DataFrame({'recency': recency, 'frequency': frequency, 'monetary': monetary})
```

```
def segment_customer(row):
    if row['recency'].year >= 2012 and row['frequency'] >= 15 and row['monetary'] > 1000:
        return 'P0'
    elif (2011 <= row['recency'].year < 2012) and (10 < row['frequency'] <= 15) and (500 < row['monetary'] <= 1000):
        return 'P1'
    else:
        return 'P2'

rfm['Segment'] = rfm.apply(segment_customer, axis=1)
```

```
rfm
```

customer_id	recency	frequency	monetary	Segment
CS1112	2015-01-14	15	1012	P0
CS1113	2015-02-09	20	1490	P0
CS1114	2015-02-12	19	1432	P0
CS1115	2015-03-05	22	1659	P0
CS1116	2014-08-25	13	857	P2
...	...	...	...	...
CS8996	2014-12-09	13	582	P2
CS8997	2014-06-28	14	543	P2
CS8999	2014-07-02	12	383	P2
CS9000	2015-02-28	13	533	P2

6884 rows × 4 columns

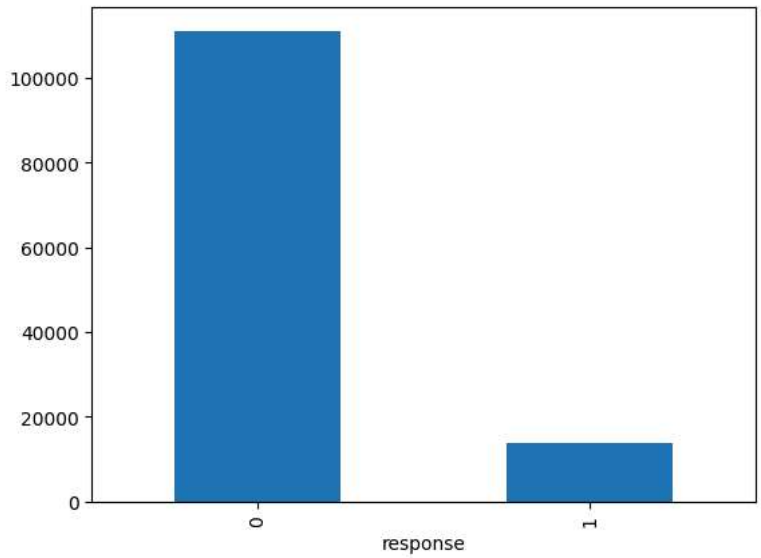
```
set(rfm['Segment'])
```

```
{'P0', 'P2'}
```

Churn Analysis

```
# Count the number of churned and active customers
churn_counts = df['response'].value_counts()

# Plot
churn_counts.plot(kind='bar')
```

```
<Axes: xlabel='response'>
```



Analyzing top customers

```
# Top 5 customers
top_5_customers = monetary.sort_values(ascending=False).head(5).index

# Filter transactions of top 5 customers
```